

Arkadiy Pechnikov

Tech Lead | Python | High-Load Systems & Distributed Architecture

Remote / Open to Relocation | arkptz@gmail.com | github.com/Arkptz | linkedin.com/in/arkptz | t.me/arkptz

PROFESSIONAL SUMMARY

Engineering leader who still writes code. Built a 9-person team from zero and delivered a real-time data platform on \$100K - roughly a tenth of what competitors spent. Architected an automated API client pipeline (mitmproxy → OpenAPI → SDK generation) that cut third-party integration from ~1 month to ~1 week across 20+ undocumented APIs. Migrated a failing monolith to Temporal.io, pushing throughput 20x to 200 jobs/sec. Comfortable hiring, managing budgets, running code reviews, and debugging protocol-level issues when the team needs it. Looking for a lead role where I can build teams and still shape the architecture.

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript/TypeScript, SQL

Leadership & Process: Team Building (up to 9), Code Review, Architecture Decisions, Hiring, Budget Management, Agile

Frameworks & Libraries: FastAPI, aiohttp, asyncio, Celery, Temporal.io, Pydantic, SQLAlchemy

Cloud & DevOps: Kubernetes, Docker, FluxCD, GitHub Actions, CI/CD, Helm, Cloudflare, MinIO, NixOS, k3s, Terraform

Databases: PostgreSQL, Redis, MySQL, Elasticsearch, TimescaleDB

Observability: Grafana, Victoria Metrics, Loki, Sentry

PROFESSIONAL EXPERIENCE

Lead Backend Engineer

[Fenixwb - Wildberries slot booking service - 200 jobs/sec, 10K+ users | Oct 2024 - Sep 2025 | Remote](#)

- Inherited an unstable monolith prototype pushing ~10 tasks/sec - redesigned the entire backend on Temporal.io with ~10 microservices integrated with Elasticsearch and MySQL
- Pushed throughput from 10 to 200 background jobs/sec (20x), the threshold that let the product handle real user load and start onboarding paying customers
- Disassembled a WASM captcha binary in Ghidra, then rewrote the solver in Rust - execution dropped from 2s to 0.1s (20x faster), giving clients higher booking success rates than competing bots
- Migrated from MongoDB to PostgreSQL - simpler queries, fewer ops headaches for a 3-person backend team
- Sourced and hired 2 engineers (backend + DevOps) from scratch, building the engineering function from zero - owned architecture calls, task allocation, and code review
- Went from zero monitoring to production-grade observability: Grafana + Victoria Metrics + Loki with custom dashboards and alerting - incidents detected in ~3 minutes vs previously hours, MTTR dropped from days to hours
- Chose to scrap the failing monolith for Temporal.io over incremental patches - the architectural call that enabled 10→200 tasks/sec across ~10 services

Tech: Rust, Python, Temporal.io, Elasticsearch, MySQL, PostgreSQL, asyncio, Grafana, Victoria Metrics, Loki, Docker, Kubernetes

Technical Lead

[Cryptosofters - Distributed automation platform with visual workflow builder - near-production, drag-and-drop operation pipelines | May 2023 - Nov 2024 | Remote](#)

- Designed distributed automation platform with drag-and-drop workflow constructor - users configured chains of multi-step operations without code. Built to near-production before investment stopped
- Under the hood: event-driven state machine with per-entity operation queues, automatic retry with exponential backoff, and operation rollback on partial failures - the engine that made drag-and-drop orchestration reliable enough for production use
- Zero security incidents across the platform's lifetime. Architected per-user isolation, credential management, and operation signing in a multi-tenant environment where investor confidence hinged on it
- Reverse-engineered undocumented third-party services via traffic analysis and protocol inspection - often the only way to integrate systems with no SDK, no docs, and no public API
- Architected the automated API client pipeline: mitmproxy traffic capture → OpenAPI spec → type-safe Python SDK for 20+ undocumented third-party APIs. Cut integration from ~1 month to ~1 week (75%) - the force multiplier that made a 9-person team viable on a \$100K budget
- Integrated 20+ external service APIs with a unified interface - per-integration time dropped from ~1 week to ~1 day (~80%), and any team member could add services, not just the specialists

- 9 engineers (4 backend, 3 frontend, 1 specialist, 1 PM), \$100K annual budget, zero turnover - kept the team together through a budget crisis that would've scattered most startups
- Grew the team from 0 to 9: wrote job descriptions, ran technical interviews, onboarded everyone, and shaped engineering culture from day one
- Set up code review standards, CI/CD pipelines, sprint cadence, and knowledge-sharing sessions - bug escape rate dropped and delivery became predictable enough to commit to external timelines
- Runway hit zero. Assembled an investor pitch deck, led the fundraising conversations myself, and bought the team more time - the moment I learned operational discipline matters as much as architecture

Tech: Python, mitmproxy, OpenAPI, asyncio, PostgreSQL, Redis, Docker

Python Developer / Tech Lead

Vitrinagram - Telegram MiniApp e-commerce platform - multi-tenant storefronts, 1,000+ concurrent users per store | Nov 2021 - Apr 2023 | Remote (Moscow-based)

- Built multi-tenant Telegram MiniApp backends handling 1,000+ concurrent users per store - horizontal scaling via Celery + Kubernetes meant cost grew linearly with tenants, not exponentially
- Automated tenant provisioning: FluxCD + Cloudflare API pipeline, ~3 minutes from commit to live subdomain
- Per-tenant observability out of the box: Grafana + Victoria Metrics + Loki + Sentry dashboards clients could use for self-service debugging - ~40% fewer support tickets once clients could self-debug
- Led backend development across a team of 7, owning architecture decisions for multi-tenant scalability, setting technical direction for the platform, and mentoring mid-level engineers through code review and design discussions that improved team velocity over two quarters
- Spotted a technical leadership vacuum that was slowing delivery - escalated to the CEO, which led to team restructuring and a noticeable improvement in sprint completion rates

Tech: Python, FastAPI, Celery, PostgreSQL, Redis, Docker, Kubernetes, FluxCD, Grafana, Victoria Metrics, Loki, Sentry, MinIO, Cloudflare

Python Backend Developer / Lead

Freelance - High-throughput data processing, infrastructure automation, and backend systems | Jan 2018 - Present

- Built distributed task orchestration system for a client with multiple investors: 2,000+ concurrent entities, 5,000+ operations/day, fully autonomous - equivalent of eliminating a full-time ops person
- Led 3 Go developers building a low-latency data processing system with sub-100ms response times - cross-language coordination (Python orchestration, Go hot path) where milliseconds determined outcomes
- Deployed single-node k3s on NixOS with declarative config, Terraform provisioning, and GitOps workflow - reproducible infra, entire cluster from zero in under an hour
- Provisioned Kubernetes clusters on cloud platforms using Terraform for multiple clients - standardized the infra setup, cut provisioning time across engagements
- 100% completion rate across 8+ freelance engagements. Repeat clients. Handled everything end-to-end - from initial scoping call to production handoff

Tech: Python, Rust, JS/TS, PostgreSQL, Redis, asyncio, NixOS, k3s, Terraform, TimescaleDB, Docker

PROFESSIONAL DEVELOPMENT

Machine Learning Specialization - Stanford University (via Coursera) (2026)

Reinforcement Learning for Trading Strategies - New York Institute of Finance (2026)

Using Machine Learning in Trading and Finance - New York Institute of Finance (2026)

LANGUAGES

Russian: Native | English: Upper-Intermediate (B2)